

ESB-R ECO-GREEN BIOCHAR50

50 kg Input/Batch Manual Biochar Plant

A compact and efficient solution for converting biomass into high-value biochar, bio-oil and syngas.



CONVERTS
BIOMASS RESIDUES
INTO VALUABLE
BIOCHAR



BATCH-TYPE
MANUAL
OPERATION



OPERATING
TEMPERATURE
UP TO
550–650°C



ELECTRICAL
HEATING
SYSTEM



INTEGRATED
PAN FILTER
FOR SEMI-GAS
CLEANING



COMPACT &
SPACE-EFFICIENT
DESIGN

PROCESS OUTPUTS



BIOCHAR

Carbon-rich solid
product



BIO-OIL

Liquid by-product



SYNGAS

Gaseous by-product
for energy use
(where applicable)

IDEAL FOR



Research Institutes
& R&D Laboratories



Universities &
Academic Institutions



Agricultural Research
& Biomass Utilization Studies



Pilot-Scale Biochar Production
& Demonstration Projects



TECHNICAL SPECIFICATIONS

ESB-R Eco-Green Biochar50 | 50 kg Input/Batch Manual Biochar Plant



Sr. No.	Parameter	Specification
1	Make	Enersol Biopower®
2	Model	ESB-R Eco-Green Biochar50
3	Plant Type	Batch-Type Biochar Plant
4	Operation	Manual / Batch Operation
5	Biomass Input Capacity	Approx. 40–50 kg per batch
6	Feedstock	Biomass, agricultural residues and other suitable carbonaceous materials
7	Preferred Moisture	Below 15%
8	Feeding System	Manual
9	Feeding Arrangement	Openable top feeding system
10	Reactor Body Material	Stainless Steel 310 (SS 310)
11	Reactor Top Construction	Flanged, openable top arrangement
12	Reactor Insulation	High-temperature thermal insulation
13	Operating Temperature	Up to 550–650°C
14	Heating Source	Electrical Heating System
15	Power Requirement	Approx. 15 kW (≈20 HP) electrical connected load
16	Reactor Length	Approx. 48 inches
17	Reactor Diameter	Approx. 24 inches
18	Overall Machine Dimensions	Approx. L 9 ft × W 6 ft × H 5 ft
19	Approx. Machine Weight	Approx. 800 kg
20	Syngas Filtration	Semi-filtration using patented PAN filter system
21	Biochar Output	Approx. 8–10 kg per batch
22	Bio-Oil Output	Generated as a by-product, quantity varies with feedstock and operating conditions
23	Syngas Output	Generated as a by-product and may be utilized as heating fuel where selected heating configuration permits
24	Process	Controlled thermal conversion / pyrolysis under oxygen-limited conditions
25	Residence Time	Dependent on feedstock type, moisture, particle size and selected operating temperature
26	Operating Control	Manual monitoring and control

PROCESS OUTPUTS



BIOCHAR

Carbon-rich solid product



BIO-OIL

Liquid by-product



SYNGAS

Gaseous by-product (for energy use where applicable)



Note: Actual input and output may vary depending on feedstock characteristics, moisture, particle size, residence time and operating temperature.

KEY FEATURES



SS 310 Reactor

Robust construction for high-temperature operation



Manual Batch Operation

Simple and reliable operation



Operating Temperature

Up to 550-650°C



Electrical Heating System

Approx. 15 kW connected load



PAN Filter System

Semi-filtration of generated syngas



Temperature Monitoring

Thermocouple provided



Compact Design

Space-efficient layout



Multi-Feedstock Capability

Suitable for various biomass and agricultural residues

APPLICATIONS



RESEARCH & DEVELOPMENT

Ideal for laboratories and R&D institutions



AGRICULTURAL RESEARCH

Study biomass residues and soil applications



UNIVERSITIES & ACADEMIC INSTITUTIONS

For experimental and demonstration purposes



PILOT-SCALE BIOCHAR PRODUCTION

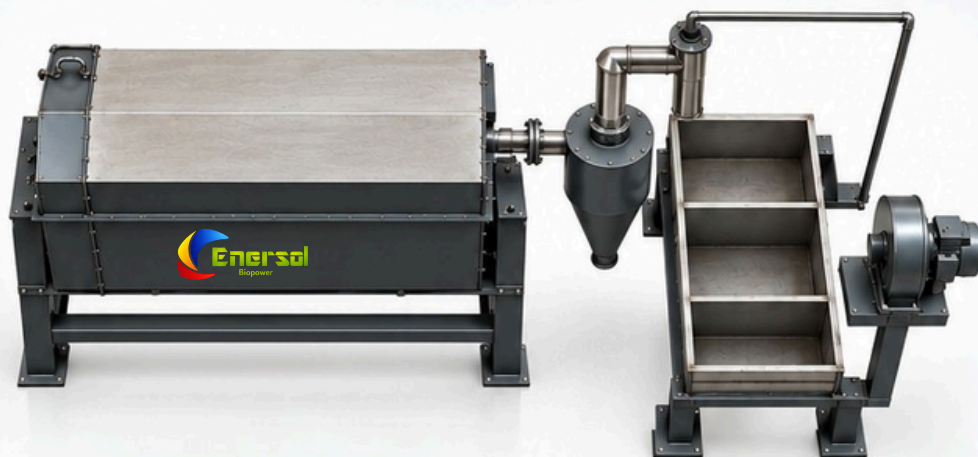
Evaluate different feedstocks and operating conditions



BIOENERGY & SUSTAINABILITY STUDIES

Support biomass-to-biochar research and circular economy initiatives

PROCESS FLOW



Reactor (Batch Type)

Manual top feeding, SS 310, insulated



Syngas Outlet

To PAN filter



PAN Filter System

Semi-filtration of syngas



Oil Collection & Cooling Tank

Collection of bio-oil by-product



Blower / Exhaust System

Supports gas flow through system

OUR KEY CUSTOMERS



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Where Waste
Becomes Power

